Jiahua Duojie

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PROFILE

Data Engineer with extensive experience in building scalable data pipelines and cloud infrastructure. Proven expertise in designing and maintaining data warehouses, data lakes, and data pipelines using PySpark, Delta Lake, and Databricks on Azure, particularly within regulated and enterprise-scale environments. Experienced in event streaming technologies, ETL processes, and DevOps practices ensuring reliable data solutions. Combines deep technical knowledge with strong communication skills to collaborate effectively with cross-functional teams. Passionate about about bridging data engineering and machine learning to create impactful, real-world applications.

CORE COMPETENCIES

- PySpark, Delta Lake, Apache Spark
- Databricks, Azure Data Factory, Azure Synapse
- Python (strong OOP and functional background)
- ETL/ELT design for regulated/financial environments
- Big Data Architecture & Workflow Orchestration (Airflow, Kafka)
- SQL, PostGIS, MongoDB, Azure Blob Storage
- Geospatial data processing, visualization, and analytics
- Agile team collaboration, solution architecture, and stakeholder communication

EDUCATION

Master of Data Science (Expected June 2026)

Lucerne University of Applied Sciences and Arts, Lucerne
Relevant Coursework: Machine Learning, Deep Learning, Human-Centered Design, Statistics, Time Serious Analysis, Data engineering, design and experiment, Data Quality

Bachelor of Computer Science (January 2017)

Nankai University, Tianjin

Languages

English (Fluent) / Chinese (Fluent) / Tibetan (Native) / German(basic)

PROFESSIONAL EXPERIENCE

Data Engineer – Qinghai Big Data Co., Ltd (2020–2023)

- Architected and implemented data pipelines for IoT sensor-based data collection, processing, and storage
- Designed and optimized ETL processes using SQL, Python, and Apache Airflow for automated data workflows

- Built and maintained cloud-based data lake infrastructure ensuring high availability and data integrity
- Implemented observability practices including monitoring, logging, and alerting to ensure reliable data solutions
- Collaborated with cross-functional teams to design data models and platforms supporting business intelligence initiativesResearch Project: Swiss Shared Mobility Analysis (Oct-Dec 2024)

Project Team Lead – Darui Media & Tech (2017–2020)

- Led development of data integration platforms connecting multiple data sources with downstream applications
- Managed deployment pipelines and release processes for smooth delivery of data solutions
- Designed and implemented PostgreSQL/PostGIS databases for spatial and analytical workloads
- Orchestrated data flows between streaming and batch systems using message queues and event-driven architecture

Web Developer – Himalaya Travel Service (2012–2017)

- Developed data collection systems and interactive mapping applications
- Built ETL processes for integrating diverse data sources into unified databases
- Implemented data validation and quality assurance processes for business-critical applications

TECHNICAL SKILLS

Cloud & Big Data: Azure, Databricks, AWS, GCP, Kubernetes, Delta Lake

Programming: Python, SQL, JavaScript, C#, FastAPI, Django

Data Platforms: Airflow, Kafka, MongoDB, PostgreSQL/PostGIS

Machine Learning: Classification, Clustering, Geospatial ML, LLMs

Visualization: Tableau, Power BI, Plotly, Kepler.gl, Matplotlib

KEY PROJECTS

Development of a Tourism Data Chatbot to Access and Interact with Complex Data

- Developed a chatbot interface for querying tourism and spatial data using natural language model
- Integrated retrieval mechanisms and interactive visualization to support cognitive accessibility
- Explored interface strategies for non-expert users engaging with structured databases

Research Project: Swiss Shared Mobility Analysis (Oct-Dec 2024)

- Developed a Databricks-powered spatial data warehouse
- Analyzed urban mobility trends through spatial clustering
- Created dashboards and geo-visualizations to support behavioral analysis
- Investigated the relationship between perceived accessibility and actual movement data

Product Classification Research for On Holding AG (Aug 2024-Jan 2025)

- Designed supervised learning models to automate product tariff code classification
- Focused on model interpretability and usability in applied business contexts
- Built preprocessing and evaluation pipelines for high-volume product data